Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $10y^2 + 12y =$

(2) Factor: $y^2 + 9y + 18 =$

(3) Factor: $16y^2 + 8y =$

(4) Factor: $6x^2 + 15x =$

(5) Factor: $x^2 + 2x + 1 =$

(6) Factor: $x^2 - 36 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 64 =$

(2) Factor: $10x^2 + 8x =$

(3) Factor: $x^2 - 9 =$

(4) Factor: $12y^2 + 4y =$

(5) Factor: $4x^2 + 12x =$

(6) Factor: $x^2 - 25 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 + 8x + 16 =$

(2) Factor: $16x^2 + 4x =$

(3) Factor: $y^2 + 8y + 12 =$

(4) Factor: $y^2 + 5y + 6 =$

(5) Factor: $x^2 - 25 =$

(6) Factor: $y^2 - 36 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 16 =$

(2) Factor: $y^2 + 7y + 12 =$

(3) Factor: $y^2 - 64 =$

(4) Factor: $14x^2 + 8x =$

(5) Factor: $24y^2 + 12y =$

(6) Factor: $24x^2 + 6x =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $28x^2 + 20x =$

(2) Factor: $x^2 - 49 =$

(3) Factor: $y^2 + 9y + 18 =$

(4) Factor: $y^2 - 16 =$

(5) Factor: $y^2 - 16 =$

(6) Factor: $y^2 + 4y + 4 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $12x^2 + 24x =$

(2) Factor: $x^2 + 6x + 8 =$

(3) Factor: $y^2 - 4 =$

(4) Factor: $28x^2 + 8x =$

(5) Factor: $x^2 - 25 =$

(6) Factor: $x^2 - 49 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 81 =$

(2) Factor: $y^2 + 6y + 9 =$

(3) Factor: $y^2 - 81 =$

(4) Factor: $16y^2 + 8y =$

(5) Factor: $y^2 - 64 =$

(6) Factor: $14x^2 + 6x =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 81 =$

(2) Factor: $20y^2 + 24y =$

(3) Factor: $6y^2 + 8y =$

(4) Factor: $x^2 - 81 =$

(5) Factor: $x^2 - 25 =$

(6) Factor: $20y^2 + 24y =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 36 =$

(2) Factor: $12x^2 + 9x =$

(3) Factor: $x^2 + 9x + 20 =$

(4) Factor: $y^2 + 7y + 12 =$

(5) Factor: $y^2 + 6y + 8 =$

(6) Factor: $y^2 + 4y + 4 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $8y^2 + 8y =$

(2) Factor: $y^2 + 4y + 3 =$

(3) Factor: $y^2 + 3y + 2 =$

(4) Factor: $12x^2 + 12x =$

(5) Factor: $y^2 + 6y + 9 =$

(6) Factor: $x^2 - 49 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 25 =$

(2) Factor: $x^2 + 9x + 18 =$

(3) Factor: $6x^2 + 15x =$

(4) Factor: $y^2 - 4 =$

(5) Factor: $x^2 + 3x + 2 =$

(6) Factor: $y^2 - 49 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 + 7y + 10 =$

(2) Factor: $x^2 + 2x + 1 =$

(3) Factor: $x^2 + 5x + 4 =$

(4) Factor: $4x^2 + 6x =$

(5) Factor: $16y^2 + 8y =$

(6) Factor: $y^2 + 5y + 4 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 + 3x + 2 =$

(2) Factor: $8y^2 + 4y =$

(3) Factor: $y^2 + 3y + 2 =$

(4) Factor: $x^2 - 25 =$

(5) Factor: $x^2 + 5x + 4 =$

(6) Factor: $x^2 - 25 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $18y^2 + 15y =$

(2) Factor: $6y^2 + 18y =$

(3) Factor: $4x^2 + 8x =$

(4) Factor: $x^2 - 64 =$

(5) Factor: $y^2 - 9 =$

(6) Factor: $12x^2 + 15x =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 16 =$

(2) Factor: $20x^2 + 20x =$

(3) Factor: $y^2 + 2y + 1 =$

(4) Factor: $24y^2 + 8y =$

(5) Factor: $y^2 + 5y + 4 =$

(6) Factor: $y^2 - 64 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 + 7y + 12 =$

(2) Factor: $x^2 + 6x + 5 =$

(3) Factor: $y^2 + 2y + 1 =$

(4) Factor: $x^2 - 9 =$

(5) Factor: $y^2 - 49 =$

(6) Factor: $24y^2 + 15y =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $12x^2 + 12x =$

(2) Factor: $y^2 + 4y + 4 =$

(3) Factor: $y^2 - 36 =$

(4) Factor: $x^2 - 9 =$

(5) Factor: $12y^2 + 15y =$

(6) Factor: $x^2 - 49 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 16 =$

(2) Factor: $y^2 - 36 =$

(3) Factor: $y^2 + 5y + 6 =$

(4) Factor: $y^2 - 64 =$

(5) Factor: $x^2 - 25 =$

(6) Factor: $x^2 + 7x + 6 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 81 =$

(2) Factor: $x^2 - 64 =$

(3) Factor: $y^2 + 8y + 12 =$

(4) Factor: $x^2 - 36 =$

(5) Factor: $x^2 + 8x + 16 =$

(6) Factor: $24y^2 + 20y =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 81 =$

(2) Factor: $x^2 + 5x + 4 =$

(3) Factor: $6y^2 + 15y =$

(4) Factor: $y^2 + 7y + 12 =$

(5) Factor: $y^2 - 16 =$

(6) Factor: $y^2 - 25 =$

Section 5 - Solutions

41a-1: $2y(5y+6)$

41a-2: $(y+3)(y+6)$

41a-3: $4y(4y+2)$

41a-4: $3x(2x+5)$

41a-5: $(x+1)(x+1)$

41a-6: $(x+6)(x-6)$

41b-1: $(x+4)(x-4)$

41b-2: $(y+3)(y+4)$

41b-3: $(y+8)(y-8)$

41b-4: $2x(7x+4)$

41b-5: $4y(6y+3)$

41b-6: $3x(8x+2)$

42a-1: $(y+8)(y-8)$

42a-2: $2x(5x+4)$

42a-3: $(x+3)(x-3)$

Section 5 - Solutions (continued)

43b-1: $(x+9)(x-9)$

43b-2: $4y(5y+6)$

43b-3: $2y(3y+4)$

43b-4: $(x+9)(x-9)$

43b-5: $(x+5)(x-5)$

43b-6: $4y(5y+6)$

44a-1: $4x(3x+6)$

44a-2: $(x+4)(x+2)$

44a-3: $(y+2)(y-2)$

44a-4: $4x(7x+2)$

44a-5: $(x+5)(x-5)$

44a-6: $(x+7)(x-7)$

44b-1: $(y+9)(y-9)$

44b-2: $(y+3)(y+3)$

44b-3: $(y+9)(y-9)$

Section 5 - Solutions (continued)

44b-4: $2y(8y+4)$

44b-5: $(y+8)(y-8)$

44b-6: $2x(7x+3)$

45a-1: $(y+6)(y-6)$

45a-2: $3x(4x+3)$

45a-3: $(x+4)(x+5)$

45a-4: $(y+3)(y+4)$

45a-5: $(y+4)(y+2)$

45a-6: $(y+2)(y+2)$

45b-1: $(y+2)(y+5)$

45b-2: $(x+1)(x+1)$

45b-3: $(x+1)(x+4)$

45b-4: $2x(2x+3)$

45b-5: $4y(4y+2)$

45b-6: $(y+4)(y+1)$

Section 5 - Solutions (continued)

42a-4: $2y(6y+2)$

42a-5: $2x(2x+6)$

42a-6: $(x+5)(x-5)$

42b-1: $(x+4)(x+4)$

42b-2: $2x(8x+2)$

42b-3: $(y+2)(y+6)$

42b-4: $(y+3)(y+2)$

42b-5: $(x+5)(x-5)$

42b-6: $(y+6)(y-6)$

43a-1: $4x(7x+5)$

43a-2: $(x+7)(x-7)$

43a-3: $(y+3)(y+6)$

43a-4: $(y+4)(y-4)$

43a-5: $(y+4)(y-4)$

43a-6: $(y+2)(y+2)$

Section 5 - Solutions (continued)

46a-1: $4y(2y+2)$

46a-2: $(y+3)(y+1)$

46a-3: $(y+2)(y+1)$

46a-4: $4x(3x+3)$

46a-5: $(y+3)(y+3)$

46a-6: $(x+7)(x-7)$

46b-1: $(y+5)(y-5)$

46b-2: $(x+3)(x+6)$

46b-3: $3x(2x+5)$

46b-4: $(y+2)(y-2)$

46b-5: $(x+2)(x+1)$

46b-6: $(y+7)(y-7)$

47a-1: $(x+1)(x+2)$

47a-2: $2y(4y+2)$

47a-3: $(y+1)(y+2)$

Section 5 - Solutions (continued)

48b-1: $(x+4)(x-4)$

48b-2: $4x(5x+5)$

48b-3: $(y+1)(y+1)$

48b-4: $4y(6y+2)$

48b-5: $(y+4)(y+1)$

48b-6: $(y+8)(y-8)$

49a-1: $2x(6x+6)$

49a-2: $(y+2)(y+2)$

49a-3: $(y+6)(y-6)$

49a-4: $(x+3)(x-3)$

49a-5: $3y(4y+5)$

49a-6: $(x+7)(x-7)$

49b-1: $(x+9)(x-9)$

49b-2: $(x+4)(x+1)$

49b-3: $3y(2y+5)$

Section 5 - Solutions (continued)

49b-4: $(y+3)(y+4)$

49b-5: $(y+4)(y-4)$

49b-6: $(y+5)(y-5)$

50a-1: $(y+4)(y-4)$

50a-2: $(y+6)(y-6)$

50a-3: $(y+2)(y+3)$

50a-4: $(y+8)(y-8)$

50a-5: $(x+5)(x-5)$

50a-6: $(x+1)(x+6)$

50b-1: $(x+9)(x-9)$

50b-2: $(x+8)(x-8)$

50b-3: $(y+2)(y+6)$

50b-4: $(x+6)(x-6)$

50b-5: $(x+4)(x+4)$

50b-6: $4y(6y+5)$

Section 5 - Solutions (continued)

47a-4: $(x+5)(x-5)$

47a-5: $(x+4)(x+1)$

47a-6: $(x+5)(x-5)$

47b-1: $(y+3)(y+4)$

47b-2: $(x+1)(x+5)$

47b-3: $(y+1)(y+1)$

47b-4: $(x+3)(x-3)$

47b-5: $(y+7)(y-7)$

47b-6: $3y(8y+5)$

48a-1: $3y(6y+5)$

48a-2: $3y(2y+6)$

48a-3: $2x(2x+4)$

48a-4: $(x+8)(x-8)$

48a-5: $(y+3)(y-3)$

48a-6: $3x(4x+5)$